MA2510 Tutorial Solution

Question 1

1. (30 points) *In this exercise, we consider two decks of cards: a "black" deck of 52 cards (containing the values $2, \dots, 10$, J, Q, K, and A, for each of the four suits[1]), and a "red" deck of 32 cards (containing only $7, \dots, 10$, J, Q, K, and A, of the four suits).*

   (i) (6 points) *We perform the following procedure: we toss a fair coin to select one of the two decks, and we then pick two cards simultaneously (i.e., at the same time), uniformly at random, from the selected deck. Compute the probability that the two cards are of the same value.*

   Let B denote the event that the black deck was selected, and let R be the event that it was the red deck instead. By assumption,

   $$\mathbb{P}(B) = \mathbb{P}(R) = \frac{1}{2},$$

   and these two events form a partition of the sample space. Denoting by S the event that the two cards are of the same value, the Law of total probability implies that

   $$\mathbb{P}(S) = \mathbb{P}(S|B) \cdot \mathbb{P}(B) + \mathbb{P}(S|R) \cdot \mathbb{P}(R).$$

   We have

   $$\mathbb{P}(S|B) = \frac{13 \cdot \binom{4}{2}}{\binom{52}{2}} = \frac{1}{17} \quad \text{and} \quad \mathbb{P}(S|R) = \frac{8 \cdot \binom{4}{2}}{\binom{32}{2}} = \frac{3}{31}.$$

   We deduce that

   $$\mathbb{P}(S) = \frac{13 \cdot \binom{4}{2}}{\binom{52}{2}} \cdot \frac{1}{2} + \frac{8 \cdot \binom{4}{2}}{\binom{32}{2}} \cdot \frac{1}{2} = \frac{41}{527}.$$

   (ii) (4 points) *We observe that the two cards are of the same value: compute the probability that the black deck was selected.*

   We now need to apply Bayes' formula:

   $$\mathbb{P}(B|S) = \frac{\mathbb{P}(S|B) \cdot \mathbb{P}(B)}{\mathbb{P}(S)} = \frac{\frac{1}{17} \cdot \frac{1}{2}}{\frac{41}{527}} = \frac{31}{82}.$$

   (iii) (4 points) *We now merge and shuffle the two decks, to produce a deck of $52 + 32 = 84$ cards. We then pick two cards simultaneously, uniformly at random, from that large deck. Compute the probability that the two cards are of the same value.*

   We observe that there are 8 cards of each of the values $7, \dots, 10$, J, Q, K, and A, but only 4 of the remaining values $2, \dots, 6$. Hence, we obtain

   $$\mathbb{P}(S) = \frac{8 \cdot \binom{8}{2} + 5 \cdot \binom{4}{2}}{\binom{84}{2}} = \frac{127}{1743}.$$

   (iv) (4 points) *We successively pick five cards from the 84-card deck, with replacement: each of them is chosen uniformly at random, and then put back into the deck, before picking the next card. Let X_B be the number of black cards that are picked, and X_R be the number of red cards. Determine the distribution of X_B, and the distribution of X_R.*

   We pick $n = 5$ cards in an independent fashion, and each card drawn is black with probability $p = \frac{52}{84} = \frac{13}{21}$, and red with probability $1 - p = \frac{8}{21}$. We deduce that X_B and X_R are Binomial$(5, p)$ and Binomial$(5, 1 - p)$ distributed, respectively.

   (v) (6 points) *Let $X = X_B - X_R$. Compute $\mathbb{E}[X]$ and $Var(X)$.*

   By linearity,

   $$\mathbb{E}[X] = \mathbb{E}[X_B - X_R] = \mathbb{E}[X_B] - \mathbb{E}[X_R] = np - n(1-p) = \frac{25}{21}$$

   (where we used the formula for the expectation of a binomially distributed random variable in the third equality).

   As for the variance, we can write $X = X_B - (5 - X_B) = 2X_B - 5$ (since $X_B + X_R = 5$. Hence,

   $$\mathrm{Var}(X) = \mathrm{Var}(2X_B - 5) = 4\mathrm{Var}(X_B) = 4np(1-p) = \frac{2080}{441} \simeq 4.72$$

   (using that X_B has binomial distribution, and the corresponding formula for the variance). Note that it is not true that $\mathrm{Var}(X) = \mathrm{Var}(X_B) + \mathrm{Var}(X_R)$, since X_B and X_R are clearly not independent.

   (vi) (6 points) *We now pick successively $n = 7000$ cards (still with replacement), and we count the number X_{JQK} of cards which are either J, Q or K (i.e., whose value belongs to $\{J, Q, K\}$). Give an estimate of $\mathbb{P}(X_{JQK} \geq 2100)$ using the Central Limit Theorem, in terms of the cumulative distribution function Φ of a random variable with the standard normal distribution.*

   Since the observations are independent, X_{JQK} has binomial distribution, with parameters $n = 7000$, and $p = \frac{24}{84} = \frac{2}{7}$. Hence,

   $$\mathbb{E}[X_{JQK}] = n \cdot p = 2000,$$

   and

   $$\mathrm{Var}(X_{JQK}) = n \cdot p(1-p) = \frac{10000}{7} \simeq 1428.6.$$

   We can then write

   $$\mathbb{P}(X_{JQK} \geq 2100) = \mathbb{P}\left(\frac{X_{JQK} - \mathbb{E}[X_{JQK}]}{\sqrt{\mathrm{Var}(X_{JQK})}} \geq \frac{2100 - 2000}{\sqrt{10000/7}} \right) \simeq \mathbb{P}(Z \geq 2.646),$$

   where $Z \sim \mathcal{N}(0, 1)$ (using the Central Limit Theorem). Hence,

   $$\mathbb{P}(X_{JQK} \geq 2100) \simeq \Phi(-2.646).$$

[1] clubs (♣), diamonds (♢), hearts (♡), and spades (♠)

Question 2

(30 points) *We consider a six-sided die, which is balanced (all integers between* 1 *and* 6 *are equally likely to appear). We roll it twice, and we assume that the outcomes of the two rolls are independent. Consider the events* $E = \{$*the two rolls produce different numbers*$\}$, $F = \{$*the second roll produces* 5$\}$, *and* $G = \{$*the sum of the two rolls is even*$\}$.

(i) (4 points) *Compute* $\mathbb{P}(E)$.
The sample space $S = \{1,2,3,4,5,6\}^2$ has cardinality $|S| = 6^2 = 36$. The event E consists of all pairs (i,j), $1 \le i,j \le 6$ with $i \neq j$. Hence, $|E| = 36 - 6 = 30$ so

$$\mathbb{P}(E) = \frac{30}{36} = \frac{5}{6}.$$

(ii) (6 points) *Are the two events* E *and* F *independent?*
By definition, we need to check whether $\mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F)$. We already know that $\mathbb{P}(E) = \frac{5}{6}$. The event F contains 6 elementary outcomes (all pairs of the form $(i,5)$, $1 \le i \le 6$), so

$$\mathbb{P}(F) = \frac{6}{36} = \frac{1}{6}.$$

Finally, we need to determine $\mathbb{P}(E \cap F)$. Clearly, $E \cap F = \{(i,5) \,:\, i = 1,2,3,4,6\}$, so

$$\mathbb{P}(E \cap F) = \frac{5}{36}.$$

We deduce that $\mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F)$, i.e. that E and F are independent.

(iii) (6 points) *Are the two events* E *and* G *independent?*
First, $\mathbb{P}(G) = \frac{18}{36} = \frac{1}{2}$. By listing all the elements in $E \cap G$, we find $|E \cap G| = 12$. Hence,

$$\mathbb{P}(E \cap G) = \frac{12}{36} = \frac{1}{3}.$$

Hence, $\mathbb{P}(E \cap G) \neq \mathbb{P}(E) \cdot \mathbb{P}(G)$ $(= \frac{5}{12})$, so E and G are not independent.

(iv) (6 points) *Compute* $\mathbb{P}(G|E)$ *and* $\mathbb{P}(G|E^c)$.
We can compute

$$\mathbb{P}(G|E) = \frac{\mathbb{P}(E \cap G)}{\mathbb{P}(E)} = \frac{1/3}{5/6} = \frac{2}{5},$$

and

$$\mathbb{P}(G|E^c) = \frac{\mathbb{P}(E^c \cap G)}{\mathbb{P}(E^c)} = \frac{\mathbb{P}(E^c)}{\mathbb{P}(E^c)} = 1$$

($E^c \cap G = E^c$, since $E^c \subseteq G$).

(v) (4 points) *Are the three events* E, F *and* G *independent?*
They are not independent, since in particular E and G are not independent (as shown earlier).

(vi) (4 points) *We now repeat the random experiment (of rolling twice the die) until we get the same number in the two rolls. We denote by* T *the number of times that the random experiment had to be performed (so that we had to roll the die* $2 \cdot T$ *times): what is the distribution of* T*? Give the probability mass function of* T.
It has geometric distribution, with parameter $p = \frac{1}{6}$ (the probability of "success", i.e. that the two rolls produce the same number): its probability mass function is

$$p_T(k) = \mathbb{P}(T = k) = p(1-p)^{k-1} \quad (k \ge 1).$$

Question 3

(20 points) *On an assembly line for laptops, keyboards are coming from four suppliers, that we call A, B, C, and D: 40% are coming from A, 25% from B, 20% from C, and 15% from D. It is estimated that among the keyboards supplied by A, B, C, and D, respectively 0.02%, 0.03%, 0.07%, and 0.06% are defective.*

(i) (4 points) *Compute the probability that a keyboard chosen at random is defective.*

We denote by D the event that such a keyboard is defective, and by E_X the event that it was produced by supplier X, for $X = A, B, C, D$. We use the Law of Total Probability to compute the probability of D:

$$\begin{aligned}\mathbb{P}(D) &= \mathbb{P}(D|E_A)\cdot\mathbb{P}(E_A) + \mathbb{P}(D|E_B)\cdot\mathbb{P}(E_B) + \mathbb{P}(D|E_C)\cdot\mathbb{P}(E_C) + \mathbb{P}(D|E_D)\cdot\mathbb{P}(E_D)\\ &= (0.0002)\cdot(0.4) + (0.0003)\cdot(0.25) + (0.0007)\cdot(0.2) + (0.0006)\cdot(0.15)\\ &= 0.000385.\end{aligned}$$

(ii) (6 points) *We pick one keyboard at random, and it happens to be defective: compute the conditional probability that it was produced by each of the four suppliers.*

We want to compute $\mathbb{P}(E_X|D)$, for each $X = A, B, C, D$, which can be done by using Bayes' formula:

$$\mathbb{P}(E_X|D) = \frac{\mathbb{P}(D|E_X)\cdot\mathbb{P}(E_X)}{\mathbb{P}(D)}.$$

We find: $\mathbb{P}(E_A|D) \simeq 0.2078$, $\mathbb{P}(E_B|D) \simeq 0.1948$, $\mathbb{P}(E_C|D) \simeq 0.3636$, and $\mathbb{P}(E_D|D) \simeq 0.2338$.

(iii) (4 points) *On a given day, $n = 10^4$ keyboards are used, and we denote by N the number of defective ones. We assume that the "states" (defective or not) of these keyboards are independent of each other. What is the distribution of N?*

Using the independence assumption, N is Binomial distributed with parameters $n = 10^4$ and $p = 0.000385$ (the probability for each keyboard to be defective).

(iv) (6 points) *Compute the expectation and the standard deviation of N.*

Using the formulas seen in class for the Binomial distribution, we find

$$\mathbb{E}[N] = n\cdot p = 3.85$$

and

$$\mathrm{Var}(N) = n\cdot p\cdot(1-p) \simeq 3.8485.$$

Hence, the standard deviation of N is

$$\mathrm{std}(N) = \sqrt{\mathrm{Var}(N)} \simeq 1.9618.$$

Question 4

(i) (4 points) *What is the conditional distribution of X given $N = 2$?*

Given $N = 2$, we toss the coin 2 successive times, in a independent fashion. We thus have $X \sim \mathrm{Bin}(2, p)$, since we perform the experiment $n = 2$ times, and p is the probability of getting "Heads" each time ("probability of success").

(ii) (8 points) *Give, in a table, the joint probability mass function of (N, X).*

The random variable N takes values in $\{1, 2, 3, 4\}$, and X in $\{0, 1, 2, 3, 4\}$. We thus need to compute $p_{(N,X)}(n, x) = \mathbb{P}(N = n, X = x)$, for all $1 \leq n \leq 4$, and $0 \leq x \leq 4$.

$X \setminus N$	1	2	3	4
0	$\frac{1-p}{4}$	$\frac{(1-p)^2}{4}$	$\frac{(1-p)^3}{4}$	$\frac{(1-p)^4}{4}$
1	$\frac{p}{4}$	$\frac{2p(1-p)}{4}$	$\frac{3p(1-p)^2}{4}$	$\frac{4p(1-p)^3}{4}$
2	0	$\frac{p^2}{4}$	$\frac{3p^2(1-p)}{4}$	$\frac{6p^2(1-p)^2}{4}$
3	0	0	$\frac{p^3}{4}$	$\frac{4p^3(1-p)}{4}$
4	0	0	0	$\frac{p^4}{4}$

(iii) (4 points) *Are N and X independent? Justify.*

Intuitively, it is clear that knowing N has a strong influence on X, since we always have $X \leq N$. For instance, the events $E = \{N = 1\}$ and $F = \{X = 4\}$ are mutually exclusive, so $\mathbb{P}(E \cap F) = \mathbb{P}(\emptyset) = 0$, while $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$, so $\mathbb{P}(E) \cdot \mathbb{P}(F) > 0$. Hence,

$$\mathbb{P}(E \cap F) \neq \mathbb{P}(E) \cdot \mathbb{P}(F),$$

so E and F are not independent, which shows that the two random variables N and X are not independent (if they were, the events $\{N = n\}$ and $\{X = x\}$ would be independent for any choice of n and x).

(iv) (4 points) *Let p_X denote the probability mass function of X. Compute $p_X(2)$.*

The probability mass function p_X of X can be obtained by summing over the rows in the table that we determined in question (ii). We find

$$p_X(2) = \frac{p^2}{4} + \frac{3p^2(1-p)}{4} + \frac{6p^2(1-p)^2}{4} = \frac{10p^2 - 15p^3 + 6p^4}{4}.$$

(v) (6 points) *Find the conditional distribution of N given $X = 3$.*

We need to determine

$$p_{N|X}(n, 3) = \mathbb{P}(N = n | X = 3) = \frac{p_{(N,X)}(n, 3)}{p_X(3)} \quad (n = 0, 1, 2, 3, 4).$$

We find the following values.

- For $n = 0, 1, 2$, $p_{N|X}(n, 3) = 0$, since $p_{(N,X)}(n, 3) = 0$ in these cases.
- For $n = 3$,

$$p_{N|X}(3, 3) = \frac{p_{(N,X)}(n, 3)}{p_X(3)} = \frac{\frac{p^3}{4}}{\frac{p^3}{4} + \frac{4p^3(1-p)}{4}} = \frac{p^3}{5p^3 - 4p^4}.$$

- For $n = 4$,

$$p_{N|X}(4, 3) = \frac{p_{(N,X)}(n, 3)}{p_X(3)} = \frac{\frac{4p^3(1-p)}{4}}{\frac{p^3}{4} + \frac{4p^3(1-p)}{4}} = \frac{4p^3 - 4p^4}{5p^3 - 4p^4}.$$

(vi) (4 points) *We now consider the random variable $Z = \mathbb{1}_{X \text{ is even}}$. Among the following values for p, indicate those for which N and Z are independent: $p = 0$, $p = \frac{1}{4}$, $p = \frac{1}{2}$, $p = \frac{3}{4}$, $p = 1$. No explanation is required.*

As can be seen on the joint probability mass function found in question (ii), N and Z are independent for $p = 0$ and $p = \frac{1}{2}$, and for no other values.

Question 5

(20 points) *We consider a sample of n i.i.d. random variables $(X_1, \dots, X_n)$, each uniformly distributed on $[-\theta, \theta]$, where $\theta \in (0, +\infty)$ is an unknown parameter.*

(i) (6 points) *Compute the expectation $\mathbb{E}[X_i]$ and $Var(X_i)$, for each $1 \leq i \leq n$.*
Using formulas for the uniform distribution on the interval $[a, b]$ (here, $a = -\theta$ and $b = \theta$), we find

$$\mathbb{E}[X_i] = \frac{-\theta + \theta}{2} = 0 \quad \text{and} \quad \mathrm{Var}(X_i) = \frac{(\theta - (-\theta))^2}{12} = \frac{\theta^2}{3}.$$

(ii) (6 points) *Find an estimator $\widehat{\theta^2}$ of θ^2 using the method of moments.*
From the previous question, $\theta^2 = 3\mathrm{Var}(X_i) = 3\mathbb{E}[(X_i)^2]$. Using the natural estimator for the second moment, we obtain

$$\widehat{\theta^2} = 3 \cdot \frac{1}{n} \sum_{i=1}^{n} (X_i)^2.$$

(iii) (8 points) *Compute the bias and the mean squared error of the estimator found in the previous question.*
This estimator is unbiased, since (by linearity) $\mathbb{E}[\widehat{\theta^2}] = 3 \cdot \mathbb{E}[(X_i)^2] = 3 \cdot \frac{\theta^2}{3} = \theta^2$, so

$$\mathrm{bias}(\widehat{\theta^2}) = \mathbb{E}[\widehat{\theta^2}] - \theta^2 = 0.$$

Hence,

$$\mathrm{MSE}(\widehat{\theta^2}) = \mathrm{Var}(\widehat{\theta^2}) + (\mathrm{bias}(\widehat{\theta^2}))^2 = \mathrm{Var}(\widehat{\theta^2}).$$

We can then write

$$\mathrm{Var}(\widehat{\theta^2}) = \frac{9}{n^2} \sum_{i=1}^{n} \mathrm{Var}((X_i)^2)$$

(using the fact that the random variables $(X_i)^2$, $1 \leq i \leq n$, are independent). By definition, $\mathrm{Var}((X_i)^2) = \mathbb{E}[(X_i)^4] - (\mathbb{E}[(X_i)^2])^2$, and we already know that $\mathbb{E}[(X_i)^2] = \mathrm{Var}(X_i) = \frac{\theta^2}{3}$. On the other hand,

$$\mathbb{E}[(X_i)^4] = \frac{1}{2\theta} \int_{-\theta}^{\theta} x^4 dx = \frac{\theta^4}{5}.$$

We finally obtain that

$$\mathrm{Var}((X_i)^2) = \frac{\theta^4}{5} - \left(\frac{\theta^2}{3}\right)^2 = \frac{4}{45}\theta^4,$$

so

$$\mathrm{MSE}(\widehat{\theta^2}) = \frac{9}{n^2} \cdot n \cdot \frac{4}{45}\theta^4 = \frac{4}{5} \cdot \frac{\theta^4}{n}.$$

Question 6

(30 points) *We consider two jointly continuous random variables X and Y, with joint probability density function*

$$f_{(X,Y)}(x,y) = \frac{3}{2}xy\mathbb{1}_{x\geq 0}\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}.$$

(i) (6 points) *Compute* $\mathbb{P}(X+Y\leq 1)$.

We can start by writing

$$\begin{aligned}\mathbb{P}(X+Y\leq 1) &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{(X,Y)}(x,y)\mathbb{1}_{x+y\leq 1}dxdy \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \frac{3}{2}xy\mathbb{1}_{x\geq 0}\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 1}dxdy.\end{aligned}$$

Hence,

$$\mathbb{P}(X+Y\leq 1) = \int_0^1 \left(\int_0^{1-x}\frac{3}{2}xydy\right)dx = \int_0^1 \frac{3}{2}x\left(\int_0^{1-x} ydy\right)dx = \frac{3}{4}\int_0^1 x(1-x)^2dx.$$

We thus obtain

$$\mathbb{P}(X+Y\leq 1) = \frac{3}{4}\int_0^1 (x^3-2x^2+x)dx = \frac{3}{4}\cdot\left[\frac{x^4}{4}-2\frac{x^3}{3}+\frac{x^2}{2}\right]_0^1 dx = \frac{3}{4}\cdot\left(\frac{1}{4}-\frac{2}{3}+\frac{1}{2}\right) = \frac{1}{16}.$$

(ii) (4 points) *Are X and Y independent? Explain why.*

No they are not independent, since the set of possible values that X can take is determined by the value of Y (and conversely), due to the condition $X+Y\leq 2$.

Question 6 continued

(iii) (6 points) *Compute the marginal probability density function of X.*

The marginal probability density function f_X of X is obtained, for any given x, by integrating over all values of y:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{(X,Y)}(x,y)dy = \int_{-\infty}^{+\infty} \frac{3}{2}xy\mathbb{1}_{x\geq 0}\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}dy = \frac{3}{2}x\mathbb{1}_{x\geq 0}\int_{-\infty}^{+\infty} y\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}dy.$$

We observe that for $x > 2$, we have $\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2} = 0$ for all y, so

$$\int_{-\infty}^{+\infty} y\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}dy = 0.$$

For $0 \leq x \leq 2$,

$$\int_{-\infty}^{+\infty} y\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}dy = \int_0^{2-x} ydy = \left[\frac{1}{2}y^2\right]_0^{2-x} = \frac{1}{2}(2-x)^2.$$

As a conclusion, we find

$$f_X(x) = \frac{3}{2}x\mathbb{1}_{x\geq 0}\cdot\frac{1}{2}(2-x)^2\mathbb{1}_{x\leq 2} = \frac{3}{4}x(2-x)^2\mathbb{1}_{0\leq x\leq 2}.$$

(iv) (6 points) *Compute* $\mathbb{E}[X]$.

We use the definition of the expectation in the case of a continuous random variable, and the expression for the probability density function f_X of X found in the previous question:

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x\cdot f_X(x)dx = \int_{-\infty}^{+\infty} x\cdot\frac{3}{4}x(2-x)^2\mathbb{1}_{0\leq x\leq 2}dx = \frac{3}{4}\int_0^2 x^2(2-x)^2dx.$$

We then compute:

$$\int_0^2 x^2(2-x)^2dx = \int_0^2 (x^4-4x^3+4x^2)dx = \left[\frac{x^5}{5}-4\frac{x^4}{4}+4\frac{x^3}{3}\right]_0^2 = \frac{16}{15}.$$

Hence,

$$\mathbb{E}[X] = \frac{3}{4}\cdot\frac{16}{15} = \frac{4}{5}.$$

(v) (8 points) *Compute* $Cov(X,Y)$.

We can write

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\cdot\mathbb{E}[Y].$$

We have just computed $\mathbb{E}[X] = \frac{4}{5}$, and since X and Y play symmetric roles, we also have similarly $\mathbb{E}[Y] = \frac{4}{5}$. Hence, there only remains to compute $\mathbb{E}[XY]$, from

$$\mathbb{E}[XY] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy\cdot f_{(X,Y)}(x,y)dxdy = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy\cdot\frac{3}{2}xy\mathbb{1}_{x\geq 0}\mathbb{1}_{y\geq 0}\mathbb{1}_{x+y\leq 2}dxdy.$$

Hence,

$$\begin{aligned}\mathbb{E}[XY] &= \frac{3}{2}\int_0^2\left(\int_0^{2-x} x^2y^2dy\right)dx = \frac{3}{2}\int_0^2 x^2\left(\int_0^{2-x} y^2dy\right)dx \\ &= \frac{3}{2}\int_0^2 x^2\cdot\left[\frac{1}{3}y^3\right]_0^{2-x}dx = \frac{3}{2}\int_0^2 x^2\cdot\frac{1}{3}(2-x)^3dx = \frac{1}{2}\int_0^2 x^2(2-x)^3dx.\end{aligned}$$

We then compute

$$\int_0^2 x^2(2-x)^3dx = \int_0^2 (8x^2-12x^3+6x^4-x^5)dx = \frac{16}{15},$$

so

$$\mathbb{E}[XY] = \frac{1}{2}\cdot\frac{16}{15} = \frac{8}{15}.$$

Finally,

$$\mathrm{Cov}(X,Y) = \frac{8}{15} - \frac{4}{5}\cdot\frac{4}{5} = -\frac{8}{75}.$$

Question 7

(30 points) *Let $\theta \in \mathbb{R}$ be given. We consider a random variable X with probability density function*

$$f_X(x) = \frac{3}{(x-\theta+1)^4} \cdot \mathbb{1}_{[\theta,+\infty)}(x).$$

(i) (6 points) *Compute $\mathbb{E}[X]$.*

We have

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x \cdot f_X(x)dx = 3\int_{\theta}^{+\infty} \frac{x}{(x-\theta+1)^4}dx.$$

We then make the change of variable $u = x - \theta + 1$, which gives

$$\mathbb{E}[X] = 3\int_{1}^{+\infty} \frac{u+\theta-1}{u^4}du = 3\left[\frac{1}{2} + \frac{\theta-1}{3}\right] = \theta + \frac{1}{2}.$$

(ii) (8 points) *Compute $Var(X)$.*

We have

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x)dx = 3\int_{\theta}^{+\infty} \frac{x^2}{(x-\theta+1)^4}dx.$$

We then make the change of variable $u = x - \theta + 1$, which gives

$$\mathbb{E}[X^2] = 3\int_{1}^{+\infty} \frac{(u+\theta-1)^2}{u^4}du = 3\left[1 + \theta - 1 + \frac{(\theta-1)^2}{3}\right] = \theta^2 + \theta + 1.$$

Hence,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2 = (\theta^2+\theta+1) - \left(\theta + \frac{1}{2}\right)^2 = \frac{3}{4}.$$

(iii) (6 points) *We now consider a sample $(X_1, \ldots, X_n)$ of size $n \geq 1$: the random variables $(X_i)_{1\leq i\leq n}$ are independent, and each is continuous with the same distribution as X, i.e. with probability density function f_X. Find an estimator of θ by the method of moments.*

Denote by $\mu_1 = \mathbb{E}[X]$ the first moment of the random variable X. We know that

$$\mu_1 = \mathbb{E}[X] = \theta + \frac{1}{2}.$$

Hence,

$$\theta = \mu_1 - \frac{1}{2}.$$

As usual, we let

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n$$

be the natural estimator for the first moment. Hence, we obtain the estimator

$$\hat{\theta}_{\mathrm{MME}} = m_1 - \frac{1}{2}.$$

Question 7 continued

(iv) (4 points) *Compute the bias of the estimator found in the previous question.*
By linearity, $\mathbb{E}[m_1] = \theta + \frac{1}{2}$, so

$$\mathbb{E}\big[\hat{\theta}_{\mathrm{MME}}\big] = \left(\theta + \frac{1}{2}\right) - \frac{1}{2} = \theta.$$

The estimator is thus unbiased: its bias is 0.

(v) (6 points) *Determine the maximum-likelihood estimator $\hat{\theta}_{MLE}$ of θ. Is it biased?*
Since the random variables are assumed to be i.i.d., the likelihood function is equal to the product

$$L(\theta; X_1, \ldots, X_n) = \prod_{i=1}^{n} \left(\frac{3}{(X_i - \theta + 1)^4} \cdot \mathbb{1}_{[\theta, +\infty)}(X_i) \right).$$

For any given observations $X_1, \ldots, X_n$, this is maximized by taking the largest value $\theta > 0$ so that the indicator functions are all non-zero, i.e., so that for all $i = 1, \ldots, n$, $X_i \geq \theta$. Hence, we obtain the estimator

$$\hat{\theta}_{\mathrm{MLE}} = \min_{1 \leq i \leq n} X_i.$$

This estimator is clearly biased: indeed, for each $i = 1, \ldots, n$, $X_i > \theta$, which implies that $\hat{\theta}_{\mathrm{MLE}} > \theta$, so

$$\mathbb{E}\big[\hat{\theta}_{\mathrm{MLE}}\big] > \theta.$$

Question 8

(i) (2 points) *Give the probability density function of X.*

By definition of the uniform distribution (here, on the interval $[a, b]$, with $a = T - \delta$ and $b = T + \delta$), the probability density function of X is

$$f_X(x) = \frac{1}{2\delta} \mathbb{1}_{[T-\delta, T+\delta]}(x).$$

(ii) (4 points) *Compute the expectation $\mathbb{E}[X]$ of X.*

Using the definition of the expectation (in the case of a continuous random variable), we have

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^{+\infty} x \cdot f_X(x) dx \\
&= \frac{1}{2\delta} \int_{T-\delta}^{T+\delta} x dx \\
&= \frac{1}{2\delta} \cdot \frac{1}{2} \Big((T+\delta)^2 - (T-\delta)^2 \Big) \\
&= T.
\end{aligned}$$

Of course, one could also apply the formula seen in class for the uniform distribution.

Question 8 continued

(iii) (4 points) *Compute the variance Var(X) of X.*
First, using the formula for the expectation of a function $f(X)$ of a random variable X, here with $f(x) = x^2$, we get

$$\begin{aligned}\mathbb{E}[X^2] &= \int_{-\infty}^{+\infty} x^2 \cdot f_X(x)dx \\ &= \frac{1}{2\delta}\int_{T-\delta}^{T+\delta} x^2 dx \\ &= \frac{1}{2\delta}\cdot\frac{1}{3}\Big((T+\delta)^3 - (T-\delta)^3\Big) \\ &= T^2 + \frac{1}{3}\delta^2.\end{aligned}$$

Hence,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \left(T^2 + \frac{1}{3}\delta^2\right) - T^2 = \frac{1}{3}\delta^2.$$

Again, it would also be fine to use directly the formula seen in class.

(iv) (6 points) *We now consider a sample of n i.i.d. random variables $(X_i)_{1\le i\le n}$, each continuous with the same distribution as X, i.e. uniform on $[T-\delta, T+\delta]$. We want to estimate the two parameters $T \in \mathbb{R}$ and $\delta \in (0, +\infty)$, which are unknown. Find estimators of T and δ using the method of moments.*

Denote by $\mu_1 = \mathbb{E}[X]$ and $\mu_2 = \mathbb{E}[X^2]$ the first and second moments of the random variable X. From the previous two questions,

$$\mu_1 = \mathbb{E}[X] = T \quad \text{and} \quad \mu_2 = \mathbb{E}[X^2] = T^2 + \frac{1}{3}\delta^2.$$

By solving for T and δ, we obtain

$$T = \mu_1 \quad \text{and} \quad \delta = (3\mu_2 - 3(\mu_1)^2)^{1/2}.$$

As usual, we let

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad m_2 = \frac{1}{n}\sum_{i=1}^{n}(X_i)^2$$

be the natural estimators for the first and second moments, respectively. Hence, we have the following estimators:

$$\hat{T} = m_1 \quad \text{and} \quad \hat{\delta} = (3m_2 - 3(m_1)^2)^{1/2}.$$

Question 8 continued

(v) (6 points) *Compute the maximum-likelihood estimators $\hat{T}_{MLE}$ and $\hat{\delta}_{MLE}$, for T and δ respectively.*

Since the random variables are assumed to be i.i.d., the likelihood function is equal to the product

$$L((T,\delta);X_1,\dots,X_n)=\prod_{i=1}^{n}\left(\frac{1}{2\delta}\mathbb{1}_{[T-\delta,T+\delta]}(X_i)\right).$$

For any given $T>0$, this is maximized by taking the smallest value $\delta>0$ so that the indicator function is non-zero, i.e. so that for all $i=1,\dots,n$, $X_i\in[T-\delta,T+\delta]$. Hence, we have necessarily

$$T-\delta\leq\min_{1\leq i\leq n}X_i\quad\text{and}\quad T+\delta\geq\max_{1\leq i\leq n}X_i.$$

The best strategy is thus to place the interval so that

$$T-\delta=\min_{1\leq i\leq n}X_i\quad\text{and}\quad T+\delta=\max_{1\leq i\leq n}X_i.$$

We deduce the estimators

$$\hat{T}_{\text{MLE}}=\frac{1}{2}\left(\max_{1\leq i\leq n}X_i+\min_{1\leq i\leq n}X_i\right)\quad\text{and}\quad\hat{\delta}_{\text{MLE}}=\frac{1}{2}\left(\max_{1\leq i\leq n}X_i-\min_{1\leq i\leq n}X_i\right).$$

(vi) (8 points) *We now let $n=10000$, and we count the number N of observations, among $X_1,\dots,X_n$, that fall in the subinterval $[T+\frac{\delta}{2},T+\delta]$, i.e.*

$$N=\left|\left\{i\in\{1,\dots,n\}\ :\ X_i\in\left[T+\frac{\delta}{2},T+\delta\right]\right\}\right|.$$

Give an estimate of $\mathbb{P}(N\geq 2600)$ using the Central Limit Theorem, in terms of the cumulative distribution function Φ of a random variable with the standard normal distribution.

Since the observations are independent, N has binomial distribution, with parameters $n=10000$, and $p=\mathbb{P}(X_i\in[T+\frac{\delta}{2},T+\delta])=\frac{1}{4}$. Hence,

$$\mathbb{E}[N]=n\cdot p=2500,$$

and

$$\text{Var}(N)=n\cdot p(1-p)=1875.$$

We can then write

$$\mathbb{P}(N\geq 2600)=\mathbb{P}\left(\frac{N-\mathbb{E}[N]}{\sqrt{\text{Var}(N)}}\geq\frac{2600-2500}{\sqrt{1875}}\right)\simeq\mathbb{P}(Z\geq 2.31),$$

where $Z\sim\mathcal{N}(0,1)$ (using the Central Limit Theorem). Hence,

$$\mathbb{P}(N\geq 2600)\simeq\Phi(-2.31).$$